

Session 5

Life Tables and Computational Demographic Techniques

An Applied Lab: Building Life Tables by Hand, in a Spreadsheet, with a Generative-AI Coding Assistant — and the Way National Statistical Offices Actually Build Them

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Questions This Session Will Answer



Life expectancy at birth

“How is it actually calculated?” Every headline e0 figure — 76.4 years for Thailand, 78.4 for the US — is the output of a reproducible calculation from raw death and population data.



Insurance pricing

“Why does a premium rise with age in a precise, calculable way?” Insurers price risk directly off the same `nx` column you will build today —



Official statistics

“How does a statistics office turn death registrations into one official number?”



AI-assisted code

“How much can an AI coding assistant get wrong — and how would you catch it?”

Learning Objectives

1

Construct a life table from raw mortality data

Derive nM_x , nq_x , l_x , ndx , nL_x , T_x , and e_x step by step, for both abridged and complete life tables — using two independent worked examples.

2

Explain how life tables are built in real practice

Trace real inputs (CRVS deaths, census population) through the institutional workflows that Thailand's NSO/MOPH, the UN, and WHO actually use to publish official life tables.

3

Build a life table computationally

Implement the construction in a spreadsheet and with a generative-AI coding assistant (Python/pandas), disclosed per course policy.

4

Critically verify AI-assisted demographic code

Apply a verification checklist to catch formula errors, off-by-one age-interval bugs, and unit mistakes.

5

Connect construction to real professional practice

See how the same $nM_x \rightarrow e_x$ pipeline prices insurance, funds pensions, computes HALE, and drives population projections -- then build two further full life tables independently in the companion practice workbook.

Agenda — 180-Minute Session Structure

Segment
Welcome, driving questions & learning objectives
Block 1 — Foundations: history, who uses life tables, types, notation & the seven core columns, incl. Group Work 1
Block 2 — Two fully worked examples, real-world applications & extensions, incl. Group Work 3
Block 3— How Life Tables Are Actually Built in Practice: real data, real institutions, real pitfalls
Block 4 — Applied Lab: spreadsheet & AI-assisted construction, verification checklist
Practice exercises, summary & discussion

Why Life Tables Matter: The 20th–21st Century Mortality Revolution

73.2

Global life expectancy at birth, 2023 (years)

+8.4

Years gained globally since 1995

77.4

Projected global life expectancy by 2054 (years)

The life table is the core analytical engine behind every one of these headline numbers. Life expectancy at birth (e_0), the single most cited demographic indicator worldwide, is a derived life-table function — it cannot be computed without first building the full nqx / lx / ndx / nLx / Tx sequence. Everything you compute today — by hand, in a spreadsheet, and with AI assistance — feeds directly into figures exactly like these.

Source: United Nations, Department of Economic and Social Affairs, Population Division. (2024). World population prospects 2024: Summary of results. https://population.un.org/wpp/assets/Files/WPP2024_Summary-of-Results.pdf

BLOCK 1

Foundations of Life Table Construction

What a life table is, why it is built the way it is, and the seven columns you will compute today.

What Is a Life Table?



A hypothetical (synthetic) cohort

A life table follows an imaginary group of, typically, 100,000 people (the "radix") from birth, applying a fixed set of age-specific mortality rates as they age.



It answers one question at every age

"Given these mortality conditions, how many of the original cohort survive to exact age x , and how many years do they collectively live?"



The demographer's core mortality model

Nearly every summary mortality indicator — life expectancy, survival probabilities, healthy-life-years — is a life-table function.



Built from real observed data

Inputs are period age-specific death rates (nMx) from vital registration or census/survey data; outputs describe a synthetic, not a real, cohort unless it is a true cohort life table.

Where the Life Table Comes From

1662 Graunt (London) → 1693 Halley (Breslau) → today's CRVS-based national life tables

1662 — John Graunt's Bills of Mortality

Graunt analyzed London's weekly death records and published the first tables summarizing age patterns of death — the founding document of demography and vital statistics (Graunt, 1662).

1693 — Edmond Halley's Breslau table

Halley used real baptism and burial records from Breslau (now Wrocław) to build the first table showing, for each age, how many of a starting cohort survived — and used it to price life annuities, the direct ancestor of Block 2's actuarial application (Halley, 1693).

18th century to today

Actuarial and government use of life tables grew continuously from Halley onward; today's life tables use the identical logical structure, built from civil registration and census data through the institutional pipelines you will see in Block 2.

Who Actually Uses Life Tables?



Actuaries & insurers

Price term-life premiums and pension annuities directly off nqx and ex ; regulators require standardized statutory mortality tables built the same way (detailed in Block 2).



Public health agencies

WHO and national ministries compute healthy life expectancy (HALE) and cause-deleted life expectancy by extending the same pipeline with disability weights and cause-of-death data (Block 1).



Governments & pension systems

National statistical offices publish official life tables (Slide 37); social-security actuaries use them to set contribution rates and retirement ages (Block 2).



Demographers & planners

UN DESA, national statistical offices, and health ministries use life-table survivorship ratios to project future populations and plan hospital, school, and eldercare capacity (Block 3).

Cohort (Generation) vs. Period Life Tables

COHORT (GENERATION) LIFE TABLE

- Follows a real birth cohort across its actual lifetime, using the mortality rates that cohort experienced at each age. Requires up to 100+ years of data to complete.
- Rarely usable in real time: only fully observable long after the cohort has died out — useful for historical analysis, not current policy.

PERIOD (CURRENT) LIFE TABLE

- Applies the mortality rates observed in a single calendar period (e.g., 2023) across all ages simultaneously, as if one synthetic cohort experienced them all.
- The standard for reporting: this is what "life expectancy at birth, 2023" always means — a snapshot, not a forecast or a real cohort's outcome.

Complete vs. Abridged Life Tables

COMPLETE LIFE TABLE

- Single-year age intervals (0, 1, 2, 3, ... up to an open-ended terminal group). Needs single-year-of-age death and population data.
- Highest resolution: preferred for detailed mortality research and for countries with high-quality vital registration.

ABRIDGED LIFE TABLE

- Wider age intervals: 0, 1–4, then 5-year groups (5–9, 10–14, ...) to an open interval (e.g., 85+). Standard $n = 5$ except the first two groups.
- Practical for most countries: used when only grouped age data are available (most censuses/surveys); today's worked examples build abridged tables.

Life-Table Notation & Terminology: Full Glossary

Symbol	Name	Meaning
x, n	Exact age; interval width	x marks the start of interval $[x, x+n)$; $n=1$ for complete tables, $n=5$ for most abridged groups ($n=4$ for ages 1–4).
nMx	Age-specific death rate	nDx / nN_x — observed deaths divided by the mid-period population at risk (Slide 17).
nq_x, np_x	Probability of dying / surviving	nq_x = probability of dying before $x+n$; $np_x = 1 - nq_x$ (Slide 18).
l_x, l_0	Survivors; radix	l_x = survivors of the cohort to exact age x ; l_0 = radix, conventionally 100,000 (Slide 19).
ndx	Deaths in the interval	$ndx = l_x - l_{x+n} = l_x \times nq_x$ (Slide 20).
nL_x, T_x	Person-years lived / remaining	nL_x = years lived by the cohort in $[x, x+n)$; $T_x = \sum$ of all nL_y for $y \geq x$ (Slides 21, 23).
e_x, nax, ω	Life expectancy; separation factor; terminal age	$e_x = T_x / l_x$; nax = average years lived in the interval by those who die in it; ω = the open-ended final age group (Slides 24, 26).

Notation follows Preston, Heuveline & Guillot (2001), the convention adopted in UN DESA and WHO official life-table publications.

Find and Interpret a Real Official Life Table

Working in group– Search for the real life table, Identify the type

1

Form groups of 3–4

Each group opens ONE real official life-table source: Thailand's NSO Statistical Yearbook (nso.go.th), the WHO Global Health Observatory (who.int/data/gho), or the UN World Population Prospects data portal (population.un.org/wpp).

2

Locate three numbers

Find e_0 , e_{65} , and the nqx (or $5q_0$) for the youngest age group in your chosen table — record the exact source, country, and reference period.

3

Predict, then check

Before you build anything yourselves today, discuss: which columns do you think were computed FROM which others? You will find out for certain in the next ten slides.

4

Report back

One spokesperson per group states, in one sentence, the country/period chosen and the e_0 value found — we will return to these real numbers as we build our own tables.

How to Read Expressions Like $5q_{10}$ and nM_x

$5q_{10}$ = probability of dying between exact ages 10 and 15 ($n=5$, $x=10$)

Leading subscript = interval width

The small number written to the left (5 in $5q_{10}$) is n , the width of the age interval in years; when it is omitted entirely, a single-year interval (or the open interval) is implied.

Main subscript = starting exact age

The number to the right (10 in $5q_{10}$) is x , the exact age at which the interval begins — so $5q_{10}$ covers ages 10 up to, but not including, 15.

The same rule, every column

$5L_{10}$, $5M_{10}$, $5a_{10}$, and T_{10} all follow the identical rule — once you can read one life-table expression, you can read all seven columns fluently.

Force of Mortality $\mu(x)$: The Hazard Behind nMx

$\mu(x)$ = the instantaneous hazard of death at exact age x

Why it matters

$\mu(x)$ is the theoretical, continuous-time quantity that nMx approximates over a finite interval — every discrete life-table formula used today is a practical approximation to this underlying hazard function.

Link to nqx

Under a constant-hazard assumption within the interval, $nqx = 1 - \exp(-n \cdot \mu(x))$; the nax -based formula used in this session is an equivalent, easier-to-compute approximation of the same relationship.

Where it reappears later

$\mu(x)$ underlies parametric mortality models (e.g., Gompertz, Heligman–Pollard) used for graduation (Slide 39) and for projecting mortality beyond the age range actually observed.

The Life Table as a Computational Pipeline



Each column is computed from the one before it. This is a strictly sequential pipeline — an error in nqx propagates through every downstream column (lx , ndx , nLx , Tx , ex). This is exactly why the verification checklist later in this session matters: a single wrong formula silently corrupts the entire table.

Column 1 — nMx : Age-Specific Death Rate

$$nMx = nDx / nNx \quad (\text{deaths in age interval } x \text{ to } x+n, \text{ divided by mid-period population})$$

Your raw input, not an output

nMx comes directly from vital registration (deaths) and census / mid-year population estimates — it is observed, not derived.

Denominator is person-time at risk

Typically approximated by the mid-year population in the interval; more precise methods use person-years of exposure directly.

Notation convention

The leading subscript n denotes interval width (e.g., $5M10$ = the rate for ages 10–14); when n is omitted, a single-year or open interval is implied.

Column 2 — nq_x : Probability of Dying in the Interval

$$nq_x = (n \cdot nM_x) / (1 + (n - nax) \cdot nM_x)$$

Converts a rate into a probability

nM_x is a rate (deaths per person-year); nq_x is the probability that a person alive at exact age x dies before reaching age $x+n$.

nax is the key adjustment factor

The average years lived in the interval by those who die in it — corrects for deaths not being evenly spread across the interval (see Slide 26).

Sanity check

nq_x must lie between 0 and 1 for every age group, including the final open interval, where ${}_{\infty}q_x = 1$ by definition.

Column 3 — l_x : Survivors to Exact Age x

$$l_0 = \text{radix } (100,000)$$

$$l_{x+n} = l_x \cdot (1 - nq_x) = l_x \cdot np_x$$

Recursive, not independent

Each l_x depends on every nq_x computed at younger ages — this is why one upstream error propagates through the whole table.

Radix is arbitrary but conventional

100,000 is standard because it turns nd_x into whole, readable numbers; the choice does not affect ex , only the scale of l_x and nd_x .

$$np_x = 1 - nq_x$$

The probability of surviving the interval — a useful complementary column, often shown alongside nq_x .

Column 4 — ndx : Deaths in the Interval

$$ndx = lx - lx+n = lx \cdot nqx$$

Deaths among the synthetic cohort

ndx is how many of the original 100,000 die between exact ages x and $x+n$, under the assumed mortality schedule.

Two equivalent formulas

$lx - lx+n$ and $lx \times nqx$ must give the same answer — a useful built-in check of your spreadsheet or code.

Sums to the radix

$\sum ndx$ across every age interval, including the open interval, must equal l_0 exactly (100,000). If not, there is an error upstream.

Column 5 — nL_x : Person-Years Lived in the Interval

$$nL_x = n \cdot l_{x+n} + n_{ax} \cdot nd_x \quad (\text{age } 0: L_0 = l_1 + a_0 \cdot d_0 \quad | \quad \text{open: } L_\omega = l_\omega / M_\omega)$$

Two components

Survivors to interval-end each contribute n full years; those who died contribute only the average fraction (n_{ax}) of the interval they lived.

Age 0 needs its own formula

a_0 is very small, not $n/2$ — infant deaths cluster in the first days/weeks of life. Using $n/2$ here is a classic error (see checklist).

Open interval is different again

For the final group $\omega+$, $L_\omega = l_\omega / M_\omega$ — there is no upper bound n , so the $n \cdot l_{x+n}$ term does not apply.

nLx in Practice: an Ordinary Interval vs. the Open Interval

ORDINARY INTERVAL: AGES 10-14

$$nLx = n \cdot lx+n + nax \cdot ndx$$

- Given: $l_{10} = 97,053$, $l_{15} = 96,714$, $n = 5$, $nax = 2.5$
- $ndx = l_{10} - l_{15} = 339$
- $nLx = 5 \times 96,714 + 2.5 \times 339 = 483,570 + 848$
- **$nLx = 484,418$ person-years**

OPEN INTERVAL: AGE 85+

$$L\omega = l\omega / M\omega$$

- Given: $l_{85} = 12,063$, $M_{85+} = 0.2162$ (observed rate)
- There is no $lx+n$ and no fixed n — everyone left dies in this interval eventually.
- **$L_{85+} = 12,063 / 0.2162 = 55,796$ person-years**
- Using the ORDINARY formula here (with a guessed n) is a common construction bug — flag it in code review.

Column 6 — Tx: Total Person-Years Remaining

$$Tx = nLx + L(x+n) + L(x+2n) + \dots + L\omega = \Sigma (\text{all } nLy \text{ for } y \geq x)$$

Sum from oldest downward

Computed as a reverse cumulative sum: start at the open interval ($T\omega = L\omega$) and add nLx moving toward age 0.

Total future person-years

Tx is the total number of years the entire lx cohort still alive at age x is expected to live, combined, under the current schedule.

Common spreadsheet bug

Summing forward (top-down) instead of backward (bottom-up) — Tx must decrease as age increases, never increase.

Column 7 — e_x : Life Expectancy at Age x

$$e_x = T_x / l_x \quad (e_0 = \text{life expectancy at birth} - \text{the most widely reported mortality measure})$$

Average remaining years

For those alive at age x , under the period mortality schedule embedded in the life table — not a prediction for any individual.

e_0 vs. e_1 : a classic teaching point

e_1 is often higher than e_0 in high-infant-mortality populations, because surviving the risky first year removes that risk from the denominator.

Comparable across time and place

Because it is a single derived number, e_x is the standard basis for cross-country and cross-period mortality comparisons.

Other Life-Table-Derived Summary Measures

Median: $l_x = l_{0/2}$ | **Modal (M):** age of max ndx | **Mean age at death** = e_0

Median age at death

The exact age at which l_x first falls to half the radix — read directly off the l_x column; less sensitive than e_0 to a small number of very long lifespans.

Modal age at death (adult mode)

The age at which ndx peaks among adult ages — a measure of the 'typical' age of death, increasingly used in ageing research because it is not distorted by infant mortality or a long right tail.

Why report more than one

In low-mortality, ageing populations, e_0 , the median, and the modal age at death can diverge meaningfully — reporting only e_0 can understate how compressed mortality has become at older ages.

The Separation Factor: nax

Definition

nax is the average number of years lived in the interval $[x, x+n)$ by those who die within it — it "separates" the interval into lived and not-lived portions for the dying.

Convenience approximation: $nax \approx n/2$

Reasonable when deaths are roughly evenly spread across the interval — true for most ages 10 and above in low-mortality populations.

Age 0 is the critical exception

Infant deaths cluster very early in life. $a_0 \approx 0.1$ – 0.3 in low-mortality populations, rising toward 0.35 – 0.5 in high-mortality populations (Coale–Demeny / UN conventions).

Ages 1–4 also need adjustment

Typically $a(1-4) \approx 1.5$ (deaths concentrated toward the younger end of the interval), not the naive $n/2 = 2.0$.

Open interval ($\omega+$)

Not an " nax " in the usual sense — the open-interval L_x is computed directly as l_ω / M_ω , bypassing nax entirely.

The Life Table as a Stationary Population

Stationary population: constant annual births = l_x alive at every age, zero growth ($r=0$)

nL_x = a real population count

If 100,000 births occurred every year and everyone always experienced today's nqx schedule, nL_x is literally how many people would be alive in age group $[x, x+n)$ at any moment.

T_x = stationary population aged $x+$

This is why nL_x and T_x are used directly in real planning: they estimate demand for age-specific services (schools, maternal care, eldercare) under current mortality, holding births constant.

e_0 = mean age at death, stationary pop.

Equivalently, e_0 equals the total stationary population (T_0) divided by the number of annual births (l_0) — one of several valid interpretations of life expectancy.

How Insurers and Pension Funds Actually Use $_{nqx}$

Pure premium, 1-year term policy $\approx$ $_{nqx}(\text{age}) \times \text{sum insured}$

Life tables price mortality risk

Insurers build, or license, mortality tables essentially identical in structure to today's — often 'select' tables reflecting insured lives, adjusted for underwriting class, smoking status, and sex.

Annuities use the same table, inverted

A pension annuity's cost depends on $_{npx}$ and $_{ex}$ — how long the insurer expects to keep paying; systematically underestimating $_{ex}$ is a real solvency risk known as longevity risk.

Regulatory life tables

Regulators such as the U.S. Society of Actuaries publish standardized experience-study mortality tables that insurers must use for statutory reserve calculations (Society of Actuaries, n.d.) — the same $_{nMx} \rightarrow$ $_{ex}$ pipeline built today, at industrial scale.

From Life Tables to Health-Adjusted Life Expectancy (HALE)

$$\text{HALE} \approx \sum [nLx \times (1 - \text{disability weight})] / 10$$

The Global Burden of Disease framework

WHO and the Institute for Health Metrics and Evaluation's Global Burden of Disease project use a life-table backbone identical to today's session, then subtract years lost to disability using survey-based disability weights for hundreds of conditions (IHME, n.d.).

Cause-deleted life tables

Public health analysts ask, 'what would e_0 be if cause X (e.g., road-traffic injury, cardiovascular disease) were eliminated?' by recomputing l_x with that cause's nqx contribution removed — a direct extension of the pipeline on Slide 16.

Why this matters for policy

HALE and cause-deleted e_0 gains are the standard evidence base for prioritizing public-health interventions and for tracking Sustainable Development Goal 3 (health) targets.

Four Things Students (and Journalists) Get Wrong About Life Tables

✗ **$e_0 \neq$ how long a real birth cohort will live**

A period e_0 describes a hypothetical cohort experiencing today's mortality rates for its entire life. A baby born today will very likely live LONGER than e_0 , because mortality tends to keep improving over their lifetime.

✗ **The radix (100,000) is not a real population count**

It is an arbitrary starting number chosen purely for computational convenience — it does not mean "100,000 people were actually studied." Changing the radix rescales l_x and nd_x but leaves every ex identical.

✗ **"Life expectancy" is not a promise about any individual**

ex is an average across the synthetic cohort. Half of that cohort, by construction, will live past ex and half will not — it says nothing about one specific person's fate.

✗ **A higher e_0 does not always mean lower infant mortality**

Two populations can share similar e_0 with very different age patterns of death — one driven by high infant mortality, another by high old-age mortality. Always inspect the full nqx / ex -by-age profile, not just e_0 .

BLOCK 2

Worked Examples: Building Life Tables Step by Step

Two small, fully worked abridged life tables — one low-mortality, one high-mortality — from raw age-specific death rates through to life expectancy.

The Seven Core Formulas at a Glance

Column	Formula	One-line meaning
nM_x	nD_x / nN_x	Observed age-specific death rate — your raw input.
nq_x	$(n \cdot nM_x) / (1 + (n - nax) \cdot nM_x)$	Probability of dying in the interval, given the rate.
l_x	$l_0=100,000; l_{x+n} = l_x \cdot (1 - nq_x)$	Survivors to exact age x — recursive.
ndx	$l_x - l_{x+n} = l_x \cdot nq_x$	Deaths among the synthetic cohort in the interval.
nL_x	$n \cdot l_{x+n} + nax \cdot ndx$ (open: l_ω / M_ω)	Person-years lived in the interval.
T_x	$\sum nL_y$ for all $y \geq x$ (bottom-up)	Total person-years remaining, all survivors.
e_x	T_x / l_x	Life expectancy at exact age x .

Keep this slide open while you work through Worked Examples 1 and 2 on the next slides.

Step 0 — The Raw Input Data

Age interval	n	nMx (rate)	nax
0–1	1	0.0200	0.30
1–4	4	0.0015	1.50
5–9	5	0.0008	2.50
10–14	5	0.0007	2.50
15–19	5	0.0012	2.50
20+ (open)	—	0.0200	—

Illustrative hypothetical cohort A (not a real country) chosen so the arithmetic is easy to follow. nax uses the standard convenience values discussed on Slide 26. We will now compute every column live.

Steps 1–2 — Compute nq_x , Then l_x

Age	$nq_x = n \cdot nM_x / (1 + (n - nax) \cdot nM_x)$	l_x
0–1	$(1 \times 0.0200) / (1 + 0.70 \times 0.0200) = 0.0197$	100,000
1–4	$(4 \times 0.0015) / (1 + 2.50 \times 0.0015) = 0.0060$	98,028
5–9	$(5 \times 0.0008) / (1 + 2.50 \times 0.0008) = 0.0040$	97,442
10–14	$(5 \times 0.0007) / (1 + 2.50 \times 0.0007) = 0.0035$	97,053
15–19	$(5 \times 0.0012) / (1 + 2.50 \times 0.0012) = 0.0060$	96,714
20+	1.0000 (by definition of the open interval)	96,135

$l_0 = \text{radix} = 100,000$. Each $l_{x+n} = l_x \times (1 - nq_x)$: e.g., $l_1 = 100,000 \times (1 - 0.0197) = 98,028$.

Steps 3–4 — Compute ndx , Then nLx

Age	$ndx = l_x - l_{x+n}$	nLx
0–1	$100,000 - 98,028 = 1,972$	$L_0 = 98,028 + 0.30 \times 1,972 = 98,619$
1–4	$98,028 - 97,442 = 586$	$4 \times 97,442 + 1.5 \times 586 = 390,646$
5–9	$97,442 - 97,053 = 389$	$5 \times 97,053 + 2.5 \times 389 = 486,236$
10–14	$97,053 - 96,714 = 339$	$5 \times 96,714 + 2.5 \times 339 = 484,416$
15–19	$96,714 - 96,135 = 579$	$5 \times 96,135 + 2.5 \times 579 = 482,121$
20+	96,135 (all remaining die eventually)	$L_{\omega} = 96,135 / 0.0200 = 4,806,750$

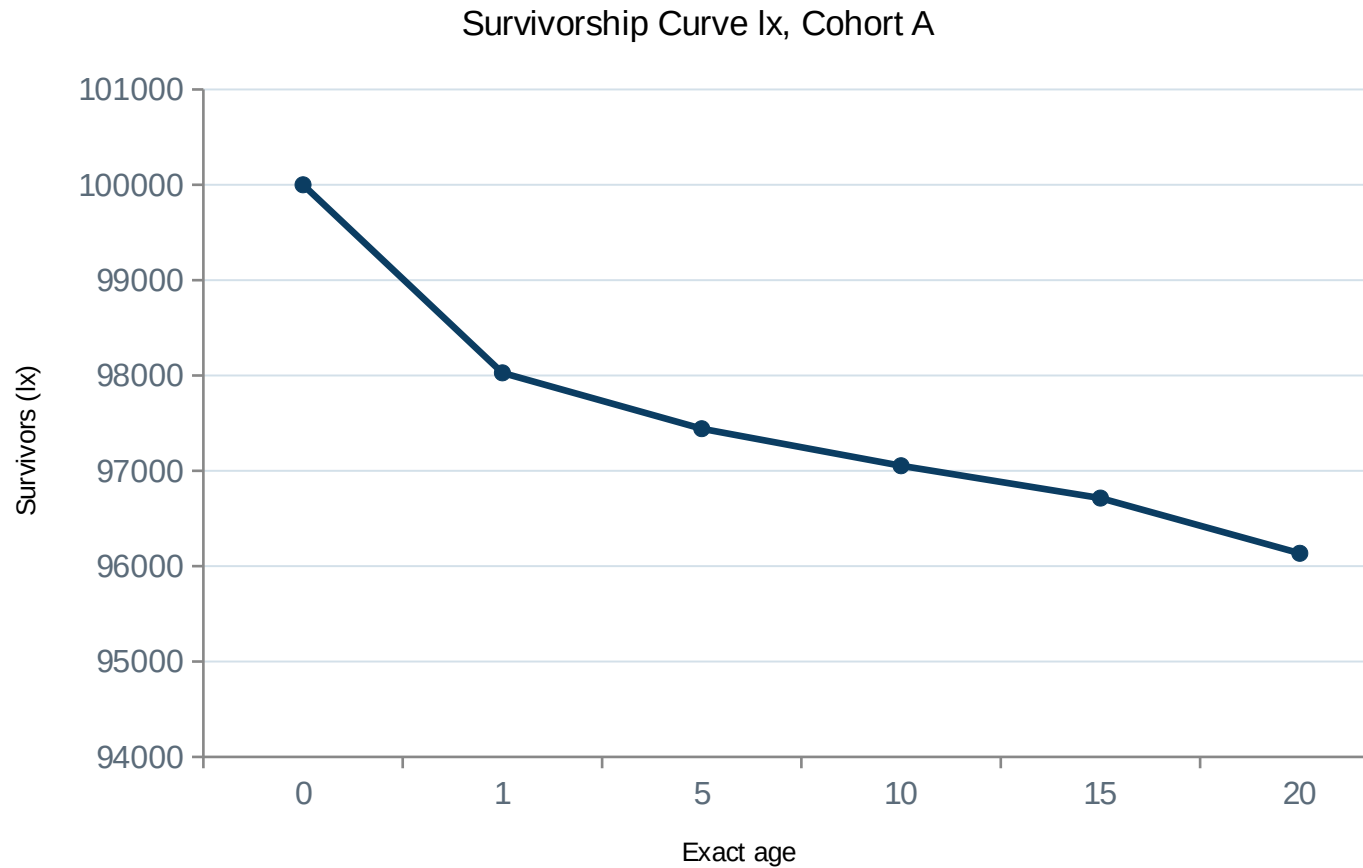
Note the age-0 special case ($L_0 = l_1 + a_0 \cdot d_0$) and the open-interval special case ($L_{\omega} = l_{\omega} / M_{\omega}$) — both differ from the general formula.

Steps 5–6 — Compute T_x , Then e_x — The Completed Table

Age	nL_x	T_x (ΣnL_x , oldest→youngest)	$e_x = T_x / l_x$
0–1	98,619	6,748,788	67.49
1–4	390,646	6,650,169	67.84
5–9	486,236	6,259,523	64.24
10–14	484,416	5,773,287	59.48
15–19	482,121	5,288,871	54.69
20+	4,806,750	4,806,750	50.00

Life expectancy at birth: $e_0 = 67.49$ years. Notice $e_1 > e_0$ — surviving infancy raises expected remaining years, a classic life-table pattern in populations with non-trivial infant mortality.

Visualizing the Result: Survivorship by Age

 **l_x declines monotonically**

By construction — the survivorship curve can never increase with age.

Steepest drop: age 0–1

Reflects the elevated infant mortality rate ($nM_0 = 0.0200$) relative to childhood ages.

 $e_x(x)$ at right

$e_0=67.5$, $e_1=67.8$, $e_5=64.2$, $e_{10}=59.5$, $e_{15}=54.7$, $e_{20}=50.0$ — declining steadily after early childhood, as expected.

From Abridged to Complete: Interpolation Methods

Why interpolate at all?

Complete (single-year) life tables give finer resolution, but many countries only publish 5-year age-grouped death and population data.

Common interpolation approaches

Reed–Merrell method, Greville's method, and osculatory interpolation split abridged nqx or nMx into single-year values while preserving the overall mortality pattern.

Model life tables as an alternative

When data are sparse or unreliable, Coale–Demeny or UN model life table families let you generate a plausible complete table from a single input, such as e_0 or $5q_0$.

Today's focus: we build the abridged table directly from data (the more common real-world task, as Block 2 showed); interpolation to single-year detail is a natural extension exercise (see Practice Exercises).

Beyond a Single Schedule: Cause-Deleted & Multistate Life Tables

Multistate table: healthy $\leftrightarrow$ disabled $\leftrightarrow$ dead, each transition its own age-specific rate

Cause-deleted life tables

Removes one cause of death's contribution to nq_x and recomputes the full $l_x \rightarrow e_x$ pipeline, answering 'how much would e_0 rise if cause X were eliminated?' — used to rank public-health priorities.

Multistate life tables

Track transitions among multiple living states, not just alive/dead — e.g., disability-free life expectancy, marital-status life tables, or working-life tables used in labor-force planning.

Same seven-column logic, more dimensions

Every one of these extensions still rests on the $nM_x \rightarrow nq_x \rightarrow l_x \rightarrow e_x$ pipeline from Slide 16; they add more transition types, not a different mathematical framework.

A Second, Higher-Mortality Dataset — Same Method, Different Numbers

Age interval	n	nMx (rate)	nax
0–1	1	0.0650	0.35
1–4	4	0.0120	1.60
5–9	5	0.0035	2.50
10–14	5	0.0030	2.50
15–19	5	0.0060	2.50
20+ (open)	—	0.0450	—

Illustrative Cohort B represents a higher-mortality population — roughly 3× the infant mortality rate and 2× the open-interval rate of Cohort A. Watching the SAME seven formulas produce a very different, much lower e_0 is the point of repeating the method on new numbers.

Steps 1–2 — Compute nq_x , Then l_x

Age	$nq_x = n \cdot nM_x / (1 + (n - nax) \cdot nM_x)$	l_x
0–1	$(1 \times 0.0650) / (1 + 0.65 \times 0.0650) = 0.0624$	100,000
1–4	$(4 \times 0.0120) / (1 + 2.40 \times 0.0120) = 0.0467$	93,763
5–9	$(5 \times 0.0035) / (1 + 2.50 \times 0.0035) = 0.0173$	89,389
10–14	$(5 \times 0.0030) / (1 + 2.50 \times 0.0030) = 0.0149$	87,838
15–19	$(5 \times 0.0060) / (1 + 2.50 \times 0.0060) = 0.0296$	86,530
20+	1.0000 (by definition of the open interval)	83,973

Compare to Cohort A: $nq(0-1)$ is more than 3× higher (0.0624 vs. 0.0197) — a realistic gap between a high- and a low-mortality population.

Steps 3–4 — Compute ndx , Then nLx

Age	$ndx = l_x - l_{x+n}$	nLx
0–1	$100,000 - 93,763 = 6,237$	$L_0 = 93,763 + 0.35 \times 6,237 = 95,946$
1–4	$93,763 - 89,389 = 4,375$	$4 \times 89,389 + 1.6 \times 4,375 = 364,555$
5–9	$89,389 - 87,838 = 1,551$	$5 \times 87,838 + 2.5 \times 1,551 = 443,067$
10–14	$87,838 - 86,530 = 1,308$	$5 \times 86,530 + 2.5 \times 1,308 = 435,921$
15–19	$86,530 - 83,973 = 2,558$	$5 \times 83,973 + 2.5 \times 2,558 = 426,258$
20+	83,973 (all remaining die eventually)	$L_{\omega} = 83,973 / 0.0450 = 1,866,062$

$\Sigma ndx = 6,237 + 4,375 + 1,551 + 1,308 + 2,558 + 83,973 = 100,002 \approx 100,000$ (rounding) — the built-in check from Slide 20 holds.

Steps 5–6 — Tx, ex, and Comparing the Two Cohorts

Age	nLx	Tx	ex
0–1	95,946	3,631,809	36.32
1–4	364,555	3,535,863	37.71
5–9	443,067	3,171,308	35.48
10–14	435,921	2,728,241	31.06
15–19	426,258	2,292,320	26.49
20+	1,866,062	1,866,062	22.22

Cohort A $e_0 = 67.49$ yrs

Cohort B $e_0 = 36.32$ yrs

Same seven formulas, same seven-column pipeline (Slide 16) — but a roughly threefold difference in infant and open-interval mortality rates translates into a 31-year gap in life expectancy at birth.

Note $e_1 > e_0$ in BOTH cohorts — the "surviving infancy raises ex" pattern from Slide 49 holds regardless of the overall mortality level.

nLx as the Engine of Population Projections and Service Planning

$$\text{Pop}(x+n, t+n) = \text{Pop}(x, t) \times [nL_{x+n} / nL_x] - \text{the survivorship ratio}$$

Every official projection uses this

UN WPP, national statistical office projections, and World Bank population forecasts all propagate each age group forward using life-table survivorship ratios nearly identical to the nL_x you built today.

Health & social-service planning

Ministries of health project future hospital-bed, maternal-care, and eldercare needs by combining population projections — which use nL_x — with age-specific service-utilization rates.

Errors compound forward in time

Because projections repeatedly apply survivorship ratios, a nax or open-interval error made today (Slide 64's checklist) does not just corrupt one year's ex — it silently biases every future projected age structure.

Comparing Countries and Tracking Progress with Life-Table Indicators

25

SDG Target 3.2 — global under-5 mortality rate target, per 1,000 live births, by 2030 (United Nations, 2015; UN Statistics Division, 2026) — directly a life-table quantity (5q0)

70

SDG Target 3.1 — global maternal mortality ratio target, per 100,000 live births, by 2030 (World Health Organization, n.d.-a)

193

UN member states each reporting comparable life-table-based mortality indicators to track these targets

5q0 (the probability of dying before age 5) and e0 are the two life-table indicators most used to compare countries and track progress toward the SDGs. Because every country's 5q0 and e0 come from the same $nMx \rightarrow nqx \rightarrow lx$ pipeline built today, they are directly comparable across very different national data systems — one reason international agencies invest heavily in standardizing the construction method (Slide 38).

Sources: United Nations. (2015). *Transforming our world: The 2030 Agenda for Sustainable Development* (A/RES/70/1). <https://sdgs.un.org/2030agenda> ; verify current progress figures against the UN SDG Indicators Database, <https://unstats.un.org/sdgs/> , before citing in coursework.

Compare Two Countries' Life-Table Indicators

1

Form new groups of 3–4

Each group selects TWO countries with contrasting income levels or regions from the UN World Population Prospects data portal (population.un.org/wpp) or the WHO Global Health Observatory (who.int/data/gho).

2

Locate e_0 and $5q_0$

Using each country's most recent published life table, record e_0 and $5q_0$ (or the closest available under-5 mortality measure) for both selected countries.

3

Discuss the gap

How large is the e_0 gap between your two countries? Which single age range (infancy, young-adult, old-age) do you suspect drives most of it, and why?

4

Report back

State the two countries, their e_0 and $5q_0$ values, your source and reference period, and the group's hypothesis about what drives the gap.

Recap: What We've Built So Far (Blocks 1–3)



The seven-column pipeline

$nMx \rightarrow nqx \rightarrow lx \rightarrow ndx \rightarrow nLx \rightarrow Tx \rightarrow ex$ — you have now built this pipeline twice, by hand, on two independently different mortality schedules.



Real construction is messier

Official life tables reconcile CRVS deaths, census population, and cause-of-death data across separate institutions — not a single clean input file.



The two classic error sites

Age 0 ($a_0 \neq n/2$) and the open interval ($L_\omega = l_\omega/M_\omega$) are where most construction bugs hide — and where the verification checklist in Block 4 focuses first.



Not just an academic exercise

Actuarial pricing, HALE, pension design, and population projections — just covered — all run on exactly this pipeline; Block 4 now turns to building it with modern tools.

BLOCK3

How Life Tables Are Actually Built in Practice

Moving from the idealized formulas of Block 1 to the real data, real institutions, and real messiness behind every published life table.

Where the Real Input Data Actually Comes From: Thailand

Deaths: civil registration (CRVS)

Death registration is the responsibility of the Ministry of Interior's Bureau of Registration Administration (BORA), Department of Local Administration — deaths must be registered within 24 hours, free of charge.

Vital statistics: Ministry of Public Health

MOPH compiles cause-specific mortality statistics using MOI registration data plus health-sector records; the two ministries have collaborated on data-matching since 1996.

Population: census & intercensal estimates

The National Statistical Office (NSO) conducts the population census and produces intercensal population estimates and projections used as the nNx denominator.

Thailand's civil registration was already close to complete by the start of the UN CRVS Decade ($\approx 98\%$ of births and deaths registered); by 2024, essentially all deaths occurring within the health sector are documented. Remaining gaps concentrate in deaths occurring outside health facilities, which is why Thailand has run dedicated cause-of-death coding and medical-training programs to close the gap.

Sources: UNESCAP CRVS Regional Hub, Thailand — civil registration and vital statistics. <https://crvs.unescap.org/country/thailand> ; Get Every One in the Picture (ESCAP). Thailand. <https://getinthepicture.org/country/thailand>

Thailand: Life Expectancy and the Persistent Sex Gap

Overall e_0 = 76.4 years (2023)

Thailand's national life table places overall life expectancy at birth at approximately 76.4 years in 2023, continuing a decades-long rise.

Large, persistent sex differential

Female life expectancy (80.9 years) exceeds male life expectancy (72.2 years) by roughly 8.6 years — driven heavily by higher male mortality from injury, cardiovascular disease, and alcohol-related causes at working ages.

Life-table implication

A rigorous national life table should always be constructed separately by sex — pooling sexes into a single nMx masks this structurally important gap, echoing the CRVS-workflow point from Slide 37.

8.6 yrs

Female – male life-expectancy gap, Thailand, 2023 (80.9 vs 72.2 years)

Source: World Bank. (2024). Life expectancy at birth, total/male/female (years) – Thailand [Data set]. World Bank Open Data, compiled from UN World Population Prospects. <https://data.worldbank.org/indicator/SP.DYN.LE00.IN?locations=TH>

The Practical Problems Analysts Actually Face

1

Age misreporting & heaping at older ages

In populations with historically weaker birth registration, reported ages cluster on round numbers (60, 70, 80, 90, 100). This distorts the age pattern of nMx precisely where the open interval begins.

2

Under-registration of deaths and births

Even "near-complete" CRVS systems miss some events, especially in remote areas or among marginalized groups — undercounting deaths deflates nMx and inflates apparent life expectancy.

3

Small-number instability

For small populations, subnational areas, or the oldest-old open interval, a handful of extra or missing deaths can swing nMx dramatically — rates computed on small denominators are noisy, not necessarily wrong.

4

The need to smooth rates

Because raw age-specific rates from real, imperfect data are jagged rather than smooth, analysts routinely apply graduation techniques before using nMx in a published life table (see Slide 39).

Life Tables and the Design of Pension & Social Security Systems

Contribution rate depends directly on ex at the statutory retirement age

Setting and adjusting retirement ages

Several countries index the statutory pension age to projected life expectancy, using national life-table projections much like the WPP-based tables discussed on Slide 38.

Funding sustainability

Social security actuaries project system solvency using cohort life tables extended decades into the future — a small error in assumed future nqx compounds into large funding gaps.

A genuinely high-stakes application

Because pension liabilities are computed decades in advance, the same construction errors flagged in Block 4's verification checklist (Slide 64) can, at national scale, translate directly into billions of currency units of mis-estimated liability.

How Thailand's Life Tables Actually Get Built

1. Collect

BORA (Ministry of Interior) registers deaths and civil events nationwide; NSO conducts the census and produces intercensal population estimates.

2. Compile & reconcile

MOPH matches MOI registration records against health-sector records (since 1996) to improve completeness and cause-of-death coding, especially for deaths outside health facilities.

3. Compute & publish

NSO and MOPH compile age-specific death rates and construct official abridged life tables, published in outlets such as the Statistical Yearbook Thailand.

This is a genuine multi-agency pipeline, not a single spreadsheet: registration (MOI), population (NSO), and health/cause-of-death data (MOPH) are produced by three different institutions and must be reconciled before a national life table can be built — exactly the kind of real-world data-integration step the toy examples in this session skip over.

Sources: UNESCAP CRVS Regional Hub, Thailand. <https://crvs.unescap.org/country/thailand> ; National Statistical Office Thailand, Statistical Yearbook. <https://www.nso.go.th/public/e-book/Statistical-Yearbook/>

How International Agencies Build Theirs

UN World Population Prospects

The UN Population Division combines every country's vital registration, censuses, and surveys into a single harmonized nMx / nax series for all 237 countries and areas, revised every 2 years.

WHO Global Health Observatory

WHO builds abridged life tables for 183 Member States each year, drawing on WPP population estimates, death registration reported to WHO, and Inter-agency child-mortality estimates for ages under 15.

US Census Bureau International Database & NCHS

For the US itself, NCHS builds official life tables from final death certificates, July 1 population estimates, AND supplementary Medicare beneficiary data for ages 66–99 — a real example of graduation using an independent old-age data source (see Slide 41).

Sources: UN DESA Population Division, *World Population Prospects 2024*. <https://population.un.org/wpp/> ; WHO, *WHO methods and data sources for life tables 2000–2021*. <https://www.who.int/publications/m/item/who-methods-and-data-sources-for-life-tables-2000-2021> ; Arias, E., Xu, J., & Kochanek, K. (2025). *United States life tables, 2023*. *National Vital Statistics Reports*, 74(6). <https://www.ncbi.nlm.nih.gov/books/NBK620699/>

Graduation and Smoothing: Fixing Jagged Real-World Rates

Why raw rates need graduation

Real nMx computed from actual small-area or single-year data is rarely smooth — sampling noise, age heaping, and small denominators produce a jagged, biologically implausible age pattern.

Classic graduation techniques

Analysts fit the raw nMx / nqx sequence to a smooth mathematical curve or reference pattern — e.g., relational logit models, splines, or a Heligman–Pollard mortality curve — before publication.

Model life tables as a related tool

When data are sparse or unreliable, Coale–Demeny or UN model life table families let analysts generate a plausible complete table from a single robust input, such as e_0 or $5q_0$.

See also: Preston, S. H., Heuveline, P., & Guillot, M. (2001). *Demography: Measuring and modeling population processes*. Blackwell Publishers.

Real nax Methods: Beyond the Naive $n/2$ Default

Method	Used for	Key idea
Naive $n/2$	Teaching / quick estimates	Assumes deaths spread evenly across the interval — wrong at age 0 and in the open interval.
Coale–Demeny regression	Age 0 and ages 1–4, by mortality level	a_0 and a_{1-4} estimated from regression equations tied to the observed infant mortality rate (IMR).
Andreev–Kingkade method	Age 0, adopted by the Human Mortality Database	Modifies Coale–Demeny for today's very-low-mortality settings, where infant deaths concentrate even more in the first days of life.
Open-interval formula	The final age group (e.g., 85+, 100+)	No nax at all — $L_w = l_w / M_w$ is used directly, sidestepping the separation-factor concept entirely.

Sources: Andreev, K., & Kingkade, W. (2015). Average age at death in infancy and infant mortality level. *Demographic Research*, 33(13), 363–390 ; Coale, A. J., & Demeny, P. (1983). *Regional model life tables and stable populations* (2nd ed.). Academic Press.

Reading an Actual Published Life Table: US Life Tables, 2023

Published July 15, 2025 by NCHS (Arias, Xu & Kochanek), using 2023 final death counts, July 1, 2023 population estimates, and Medicare data for ages 66–99 to sharpen old-age mortality — a real, current, three-source construction exactly like the pipeline on Slide 37.

e0 (total) = 78.4 years (males 75.8, females 81.1) — up 0.9 year from 77.5 in 2022, a post-pandemic rebound.

Age x	lx (of 100,000)	What it means
0	100,000	Radix
1 (female)	99,487	99.5% of female babies survive infancy
20 (male)	98,705	$T_x(20) = 5,598,276 \rightarrow e_{20}(\text{male}) = 56.7 \text{ yrs}$
80 (total)	57,812	57.8% of the cohort survives to age 80
85 (total)	42,699	$P(\text{survive } 20 \rightarrow 85) = 42,699/98,901 = 43.2\%$

Source: Arias, E., Xu, J., & Kochanek, K. D. (2025). United States life tables, 2023. National Vital Statistics Reports, 74(6). National Center for Health Statistics. <https://www.ncbi.nlm.nih.gov/books/NBK620699/>

Real Pitfalls Analysts Encounter



Registration completeness gaps

Even countries with strong CRVS systems have pockets of under-registration — remote regions, deaths outside health facilities, undocumented populations — that bias nMx downward exactly where it is hardest to detect.



COVID-era data disruptions

US life expectancy fell 1.86 years in 2020 and a further 0.55 years in 2021 relative to 2019 — the largest two-year drop in decades — then partially rebounded to 78.4 years by 2023, illustrating how fast real mortality shocks move through the whole life-table pipeline.



Why life tables get revised

Every new round of a country's census, an updated CRVS completeness estimate, or a change in population-estimate methodology (e.g., a new census benchmark) triggers a full life-table revision — the numbers you cite today may already be superseded next year.



Small-area and subnational instability

A provincial or district-level life table built on the same formulas as a national one can still be unusable if annual deaths in some age groups number in the single digits — the formulas are correct, but the data cannot support them.

Diagnose a Data-Quality Problem in Real Mortality Data

1

Open the Session 5 Excel workbook

Use the real 'Data (Thailand 2024)' sheet — UN World Population Prospects / PopulationPyramid.net figures — already loaded in your practice workbook.

2

Inspect the oldest age groups

Compare the population and death counts for 90–94, 95–99, and 100+ — are they large enough to trust the resulting nM_x , or is this small-number territory (Slide 35)?

3

Propose a fix

As a group, choose ONE remedy discussed today — graduation/smoothing, a model life table, or an alternative nax method (Slides 39–40) — and explain why you chose it.

4

Report back

State which age group you inspected, what problem you found, and your proposed fix, in two sentences.

BLOCK 4

Computation Tools: Spreadsheet & Generative-AI Assistant

Building the same life table two more ways — and learning to catch the errors a coding assistant can silently introduce.

Building the Table in a Spreadsheet

Column	Cell logic (row i, previous row i-1)
nqx (B)	$=(n_i * M_i) / (1 + (n_i - a_i) * M_i)$
lx (C)	Row 1: =100000 Row i>1: $=C[i-1] * (1 - B[i-1])$
ndx (D)	$=C[i] * B[i] \text{ (or } =C[i] - C[i+1])$
nLx (E)	General: $=n_i * C[i+1] + a_i * D[i]$ Age 0: $=C[2] + a_0 * D[1]$ Open: $=C[\text{last}] / M[\text{last}]$
Tx (F)	Fill from the LAST row upward: $F[\text{last}] = E[\text{last}]$; $F[i] = F[i+1] + E[i]$
ex (G)	$=F[i] / C[i]$

Build column by column, left to right — and always compute Tx with a bottom-up running sum, never a single SUM() over the whole range (that silently breaks the "remaining years from age x" logic).

Building the Table with a Generative-AI Coding Assistant

```
# Prompt given to the AI assistant (disclosed, per course policy):  
# "Write a Python/pandas function that builds an abridged life table  
# from a DataFrame of age, n, and nMx, using radix 100000."  
  
def life_table(df, radix=100000):  
    df['nax'] = df['n'] / 2          # WARNING: AI default - check this!  
    df['nqx'] = (df.n*df.nMx) / (1+(df.n-df.nax)*df.nMx)  
    df.loc[df.index[-1], 'nqx'] = 1.0  # open interval  
    df['lx'] = radix  
    for i in range(1, len(df)):  
        df.loc[i, 'lx'] = df.loc[i-1, 'lx']*(1-df.loc[i-1, 'nqx'])  
    df['ndx'] = df['lx']*df['nqx']  
    df['nLx'] = df.n*df.lx.shift(-1) + df.nax*df.ndx  # WARNING: open interval breaks here  
    df['Tx'] = df['nLx'][::-1].cumsum()[::-1]  
    df['ex'] = df['Tx'] / df['lx']  
    return df
```

This AI-generated function looks plausible and runs without errors — but it contains at least two seeded bugs typical of AI-generated demographic code. Find them before the next slide.

Critical Verification Checklist for AI-Generated Demographic Code

1

Wrong or generic n_{ax} default

AI assistants default to $n_{ax} = n/2$ for every age, including age 0 — always overwrite a_0 (≈ 0.3) and $a_{(1-4)}$ (≈ 1.5) by hand.

2

Off-by-one age intervals

Check that l_x uses the PREVIOUS row's n_{qx} , not the current row's — a common shift/index bug in generated loops.

3

Ignoring the open-ended age group

The final age group (e.g., 85+) has no upper bound n — $L_{\omega} = l_{\omega}/M_{\omega}$, not $n \cdot l_x + n_{ax} \cdot n_{dx}$. Confirm the code branches for this case.

4

Silent unit errors

Rates per 1,000 vs. per 1 person; ages in months vs. years — AI code rarely infers your input's units and will compute confidently with the wrong ones.

5

T_x computed top-down, not bottom-up

T_x must be a REVERSE cumulative sum (oldest to youngest); verify T_x is monotonically decreasing with age.

6

No sanity checks on outputs

Always confirm: $\sum n_{dx} = \text{radix}$; $n_{qx} \in [0,1]$ for all rows; l_x strictly non-increasing; e_x plausible for the population studied.

Course policy: any generative-AI-assisted output must be disclosed and independently, manually verified against the formulas in Block 1 before it is used or submitted.

In-Class Discussion: Where Do the Bugs Break the Table?

```
df['nax'] = df['n'] / 2                                # BUG 1: overwrites the a0=0.3, a1-4=1.5 rule
df['nLx'] = df.n*df.lx.shift(-1) + df.nax*df.ndx      # BUG 2: .shift(-1) on the LAST row is NaN -
                                                       # open-interval nLx silently becomes NaN,
                                                       # corrupting every Tx and ex above it too.
```

Discussion prompt:

- (1) Which two life-table sanity checks from the previous slide would have caught these bugs immediately?
- (2) Rewrite both lines correctly, referencing the formulas from Slides 21 and 22.

How Common Is AI-Assisted Coding in Research Right Now?

This is now mainstream research practice

A 2024 Stack Overflow developer/researcher pulse survey found academic researchers were the professional group MOST likely to use AI code assistants, at 87%.

Chatbot use among scientists is similarly high

An informal 2023 Nature reader survey found roughly 80% of respondents had used AI chatbots for work tasks, including writing and debugging code.

Why this session treats disclosure as mandatory

Given how common AI-assisted coding already is, the professional skill that matters is not whether you use it, but whether you can verify it — exactly the checklist from Slide 64.

87%

Of academic researchers surveyed report using an AI code assistant (Stack Overflow, 2024)

Sources: Stack Overflow. (2024). Developers get by with a little help from AI. <https://stackoverflow.blog/2024/05/29/developers-get-by-with-a-little-help-from-ai-stack-overflow-knows-code-assistant-pulse-survey-results/> ; Nature. (2023). How Nature readers are using ChatGPT. <https://www.nature.com/articles/d41586-023-00500-8>

The Corrected Code: Fixing Both Seeded Bugs

```
df.loc[0, 'nax']=0.3; df.loc[1, 'nax']=1.5 | nLx(open) = lw / Mw,  
not n·lx+n + nax·ndx
```

Fix 1 — restore nax by age

Overwrite the AI's blanket $n/2$ default with $a_0 \approx 0.3$ and $a(1-4) \approx 1.5$ immediately after computing nax , exactly as flagged in the verification checklist (Slide 64, item 1).

Fix 2 — branch the open interval

Compute nLx for every ordinary row with the general formula, then overwrite the LAST row with lw/Mw — never let $.shift(-1)$ silently produce NaN there (checklist item 3).

The result now passes every sanity check

$\sum ndx = radix$, $nqx \in [0,1]$, lx strictly non-increasing, and Tx strictly decreasing — the four checks from Slide 64 that the buggy version would have failed.

Lab Task: Build, Verify, Compare

1

Step 1 — Build

In pairs, build the Slide 46 (or 53) life table in a spreadsheet AND ask an AI assistant to generate the equivalent Python/pandas code.

2

Step 2 — Compare

Do the spreadsheet and the AI-generated code produce the same results? If not, use the verification checklist to locate the discrepancy.

3

Step 3 — Disclose

Record which AI tool was used, the exact prompt given, and every correction you made to its output — this disclosure is a mandatory part of the submission.

4

Step 4 — Report

Be ready to state, in one sentence, the single most important lesson from reconciling the two methods.

Recap: Tools and Verification (Block 4)



Spreadsheets and AI, together

Both can build a correct table quickly — neither should be trusted without the manual verification checklist from Slide 64; they complement, not replace, each other.



Disclosure is mandatory

Course policy requires stating which AI tool was used, the exact prompt, and every correction made — a professional norm shared by 87% of surveyed academic researchers who already use AI code assistants.



The two classic error sites, again

Age 0 (nax) and the open interval account for most construction bugs, whether the code was written by a human or an AI assistant.



Traces back to Slide 16

HALE, cause-deleted e0, pension-cost projections, and SDG monitoring are all downstream applications of the same seven-column pipeline you built by hand today.

Key Takeaways

A life table is a sequential pipeline

$nMx \rightarrow nqx \rightarrow lx \rightarrow ndx \rightarrow nLx \rightarrow Tx \rightarrow ex$ — each column depends on the ones before it, so errors propagate.

Real construction is a multi-agency process

Official life tables come from reconciling CRVS deaths, census population, and cause-of-death data across separate institutions (Slides 33–42) — not a single clean input file.

Special cases are where errors hide

Age 0 ($a_0 \approx 0.3$, not $n/2$) and the open-ended terminal age group ($L_w = l_w/M_w$) are the two most common sources of construction errors.

Spreadsheets & AI are complementary

Both can build a correct table quickly — neither should be trusted without the manual verification checklist.

Every claim traces to a real source

From Graunt's 1662 bills of mortality to today's NSO, UN DESA, and WHO databases (Slides 55–57), the numbers in this session are never invented -- they are drawn from citable, verifiable sources you can check yourself.

Every measure connects to real practice

Actuarial pricing, HALE, pension design, and population projections are not separate topics -- they are downstream applications of the exact same pipeline built by hand today (Blocks 1 and 2).

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Practice Exercises

A mix of conceptual and computational questions

Short-answer conceptual questions test your understanding of life-table logic; calculation questions require you to actually build or extend a life table.

Use the tools from today

For calculation questions, use a spreadsheet or write your own Python/pandas code — and apply the verification checklist to your own work.

Format

20 questions across the next five slides, roughly 4 per slide. The companion Excel workbook adds two further full life-table construction exercises and ten additional problems -- complete both before the next session.

Practice Exercises (1 of 5)

Q1. Explain, in your own words, why nqx and nMx are conceptually different quantities even though nqx is calculated from nMx .

Q2. Why does the age-0 interval require a special formula for both nqx (via a_0) and nLx , rather than using the general formulas that apply to other age groups?

Q3. A colleague builds a life table and finds that T_x increases with age. What is the most likely error, and how would you locate it in a spreadsheet or code?

Q4. Explain why e_1 can be higher than e_0 in a population with meaningful infant mortality.

Practice Exercises (2 of 5)

Q5. What is the difference between a cohort life table and a period life table, and why are almost all publicly reported "life expectancy" figures period measures?

Q6. Given $l_{20} = 96,135$ and $l_{25} = 95,400$ (radix 100,000), calculate $5q_{20}$ and $5d_{20}$.

Q7. Using the Cohort A worked-example life table (Slide 49), verify by hand that $\sum n d_x$ across all six age groups equals the radix (100,000). Show your work.

Q8. A life table reports $nM_0 = 0.045$ and $a_0 = 0.35$. Using $n = 1$, compute nq_0 .

Practice Exercises (3 of 5)

Q9. For an open age interval "75+" with $l_{75} = 40,000$ and $M_{75+} = 0.085$, compute L_{75+} and e_{75} .

Q10. Using the Cohort B abridged life table (Slide 54), verify that the two equivalent formulas for ndx ($lx - lx+n$ and $lx \times nqx$) give the same answer for the 5–9 age group. Show both calculations.

Q11. Using Cohort B's inputs (Slide 53), recompute $5q_{15}$ if $nM(15-19)$ were 0.0090 instead of 0.0060, holding $nax = 2.5$. How much does l_{20} change?

Q12. Using the Session 5 Excel workbook's Thailand 2024 data, explain why nqx for the "100+" open interval is set to 1.0 by definition rather than computed from the general nqx formula.

Practice Exercises (4 of 5)

Q13.

An AI coding assistant returns Python code that computes nL_x as $n \times l_x$ for every age group, including age 0. Identify the error and explain, with reference to the checklist, which slide's formula it violates.

Q14.

You ask an AI assistant to build a complete (single-year) life table from 5-year abridged data without specifying an interpolation method. What should you check before trusting its output?

Q15.

Design one additional automated sanity check (beyond those on Slide 64) that you could add to a Python life-table function to catch construction errors automatically.

Q16.

Using the Thailand example (Slide 34), explain one demographic reason a national life table should be constructed separately by sex rather than pooled.

Practice Exercises (5 of 5)

Q17.

Using the CRVS workflow on Slide 37, explain why a life table built from ONLY Ministry of Interior registration data (without the Ministry of Public Health matching step) might understate cause-specific mortality even if the total death count is accurate.

Q18.

Why does the US NCHS life table (Slide 41) supplement vital-registration data with Medicare records for ages 66–99? What real-world data problem is this solving, and how does it relate to the "small-number instability" issue from Slide 35?

Q19.

A national statistical office revises last year's official life table after this year's census. List two genuine reasons (from Slide 42) this revision might happen, and explain why it does NOT mean last year's table was "wrong."

Q20.

Compare the Andreev–Kingade nax method to the naive $n/2$ default (Slide 40). In which part of the age range does the choice of method matter most for the resulting e_0 , and why?

Questions & Discussion

Next session: Fertility, Contraception, and Reproductive Health

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